Discrete Structure (CST-002) - Unit 5

Unit 5 - Graphs & Trees - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: graph theory and trees - vertices, edges, paths, Euler and Hamilton circuits, planarity, spanning trees and tree traversals.

Key ideas: A **graph** G = (V, E). The **handshaking lemma**: sum of all degrees = 2 x number of edges (so the number of odd-degree vertices is always even). An **Euler circuit** uses every edge exactly once and exists when every vertex has even degree; a **Hamilton circuit** visits every vertex exactly once (no easy test - remember that contrast). A graph is **bipartite** if vertices split into two sets with no edge inside a set (equivalently, no odd cycle). **Planar** graphs draw without crossings; Euler's formula: v - e + r = 2. A **tree** is a connected graph with no cycles: n vertices always give n - 1 edges. A **spanning tree** touches every vertex; **Kruskal** (cheapest edge first, skip cycles) and **Prim** (grow one tree from a start vertex) find the minimum spanning tree. Binary tree traversals: **preorder** (root-left-right), **inorder** (left-root-right), **postorder** (left-right-root).

Exam pointers: Euler/Hamilton existence checks, Euler's formula numerics, one Kruskal/Prim step-by-step MST, and one traversal question appear almost every year.

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